
Null Congruence Expansion in Curvature-Extremum Spacetime

Relativistic Optical Consistency of Path-Integrated Redshift in Void-Node Geometry

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Abstract

In a companion paper we identified Mass Deficit Nodes (MDNs) as local minima of spacetime curvature organizing large cosmic voids. Additionally, we identify MDNs as geometric structures complementary to black holes under Curvature Extremum Duality. In this work we examine the optical consequences of such curvature minima within classical general relativity.

Starting from the exact general-relativistic expression for photon redshift, we analyze the evolution of null congruences traversing expansion-dominated geometry organized around curvature minima (“void nodes”) while timelike worldlines remain confined to filamentary, mass-concentrated structure. We show that when null and timelike congruences sample systematically different expansion histories, the redshift rate acquires a non-canceling, path-integrated contribution governed by congruence geometry rather than boundary-localized potential exchange.

The resulting effect is distinct from Integrated Sachs–Wolfe and Rees–Sciama contributions, persists for stationary curvature structure, and may be interpreted as a null-averaged optical bias in void-dominated spacetime. No modification of Einstein’s equations is introduced. The analysis establishes the relativistic-optical consistency of a geometric redshift contribution arising from curvature-minimum organization, providing the formal optical underpinning for the phenomenology introduced in Paper 1.

We further derive a leading-order scaling relation for this path-integration contribution, showing that its magnitude is governed by the cumulative affine path length through expansion-dominated void-node basins rather than local extrema. This formulation naturally admits a dimensionless observable that isolates line-of-sight-dependent deviations from homogeneous expectations, implying that while the mean effect may be absorbed into inferred cosmological parameters, residuals should correlate with the integrated basin structure traversed by photo paths.

1. Introduction

Cosmological redshift is most commonly interpreted as a manifestation of homogeneous metric expansion, supplemented in inhomogeneous settings by localized gravitational effects such as the Integrated Sachs–Wolfe (ISW) and Rees–Sciama corrections. These latter effects arise from

time-dependent gravitational potentials and are typically small due to near-cancellation between entry and exit contributions.

In Paper 1, we argued that this standard accounting overlooks a geometrically distinct feature of spacetime: the existence of curvature minima that organize void-dominated regions, complementary to curvature maxima that organize matter-dominated structures such as black holes and galaxies. These curvature minima, Mass Deficit Nodes (MDNs) or *void nodes*, do not merely trace underdensity but act as organizing centers of expansion-dominated geometry under Curvature Extremum Duality.

This motivates a precise relativistic-optical question:

Does general relativity permit a non-canceling, path-integrated redshift contribution arising purely from null propagation through curvature-minimum geometry, when timelike observers do not share the same expansion history?

The present paper addresses this question at the level of null congruence geometry, without parameter fitting, cosmological model replacement, or metric construction. Our aim is to determine whether such an effect is allowed in principle by the exact structure of general-relativistic redshift.

2. Null Congruences and Optical Scalars

A bundle of neighboring photon trajectories is described in general relativity by a null congruence with tangent vector

$$k^a = \frac{dx^a}{d\lambda},$$

where λ is an affine parameter.

The local evolution of a null congruence is characterized by three optical scalars:

- the expansion θ ,
- the shear σ_{ab} ,
- and the vorticity ω_{ab} .

The expansion scalar

$$\theta = \nabla_a k^a$$

measures the fractional rate of change of the congruence's cross-sectional area. Its evolution is governed by the null Raychaudhuri equation:

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{ab}\sigma^{ab} + \omega_{ab}\omega^{ab} - R_{ab}k^ak^b.$$

In curvature-maximum regions associated with mass concentrations, the Ricci focusing term $R_{ab}k^ak^b$ is typically positive and drives convergence of null geodesics. By contrast, in curvature-minimum regions organized around MDNs, Ricci focusing is suppressed, allowing $\theta > 0$ to persist over extended affine intervals. This divergence of null congruences is the fundamental optical feature of void-node geometry.

3. Redshift in General Relativity: Exact Formulation

The frequency ω of a photon measured by an observer with four-velocity u^a is given exactly by

$$\omega = -u^ak_a.$$

For a photon exchanged between an emitter and an observer, the observed redshift is therefore

$$1 + z = \frac{(k^au_a)_{\text{em}}}{(k^au_a)_{\text{obs}}}.$$

Taking the logarithmic derivative along the photon path yields the exact identity

$$\frac{d}{d\lambda} \ln(1 + z) = -\frac{1}{k^au_a} k^b \nabla_b u_a,$$

which follows directly from the geodesic equation $k^b \nabla_b k^a = 0$.

Redshift in general relativity is therefore governed by the relative evolution of the null congruence and the timelike observer congruence. No approximation has been made.

In homogeneous FLRW spacetime, both congruences probe the same expansion history, and the expression reduces to the familiar Hubble relation. In inhomogeneous spacetime, particularly in the presence of curvature-organized voids, this coincidence is not guaranteed.

4. Null–Timelike Congruence Decoupling in Void-Node Geometry

In spacetime organized by Curvature Extremum Duality, a categorical trajectory distinction arises.

- Null geodesics traverse the interiors of voids organized around curvature minima (MDNs).
- Timelike worldlines (galaxies and observers) are dynamically confined to filamentary and wall structures associated with curvature maxima.

As a result, the null congruence samples an expansion history characterized by suppressed Ricci focusing and persistent divergence, while the timelike congruence samples a materially distinct expansion history influenced by mass concentration.

This null–timelike congruence decoupling is the geometric origin of the effect analyzed here. It arises not from exotic physics, but from the complementary organization of spacetime around curvature extrema.

5. Effective Expansion Along Null Paths

Decomposing $\nabla_a u_b$ into expansion, shear, and vorticity, and contracting with the null tangent, the redshift rate may be written schematically as

$$\frac{d}{d\lambda} \ln(1+z) = \frac{1}{3} \Theta_{\text{eff}}(\lambda) + (\text{shear corrections}).$$

Here $\Theta_{\text{eff}}(\lambda)$ denotes the effective volumetric expansion sampled along the null path.

- In homogeneous spacetime, $\Theta_{\text{eff}} = 3H$ everywhere.
- In void-node-organized geometry, $\Theta_{\text{eff}}(\lambda)$ remains elevated across extended affine intervals.

Shear contributions are generically present in inhomogeneous spacetime but enter as sub-leading corrections in large, approximately isotropic void interiors. Their presence modifies the magnitude but not the path-integrated character of the effect.

6. Distinction from ISW and Rees–Sciama Effects

It is essential to distinguish the mechanism considered here from potential-based redshift effects.

- ISW and Rees–Sciama effects arise from explicit time dependence of the gravitational potential and are boundary-localized.

- In static or slowly evolving curvature structure, their contributions exhibit near-cancellation.

By contrast, the contribution derived here enters through the tensor $k^b \nabla_b u_a$ and depends on the geometric expansion of the null congruence itself. It persists even for stationary curvature minima and does not rely on temporal evolution of the potential.

The distinction is therefore structural, not interpretive.

7. Interpretation as Null-Averaged Optical Bias

The effect may be interpreted as a null-averaged optical bias inherent to void-dominated spacetime geometry.

Spatial averaging techniques weight geometry by volume on spacelike hypersurfaces. Null averaging weights geometry by affine path length along light rays. In strongly inhomogeneous spacetime, these averages are not equivalent.

Because void-node-organized regions dominate affine path length, their geometric properties disproportionately influence null observables even when global expansion remains well approximated by FLRW.

8. Scaling Behavior and an Observable Parameter for Path-Integrated Redshift

8.1 Motivation

The derivation presented in Appendix A establishes that, in inhomogeneous spacetime, redshift accumulation along null geodesics depends on the expansion history sampled by the photon path rather than solely on a homogeneous background evolution. In spacetime organized around curvature-minimum structures (Mass Deficit Nodes, or void nodes), this leads to a path-integrated contribution arising from sustained traversal of expansion-dominated regions.

While this establishes the *existence* of such a contribution, it is also useful to determine how its magnitude scales with the geometric properties of the regions traversed. The goal of this section is therefore to extract a leading-order scaling relation and to define a dimensionless observable parameter that isolates this effect from the baseline cosmological redshift.

No attempt is made to construct a detailed model or perform parameter estimation; the results below are intended only to capture the dominant geometric dependencies.

8.2 Basin-Dominated Approximation

As discussed in Paper 1, curvature-minimum nodes (MDNs) organize extended basin structures characterized by:

- suppressed matter density,
- reduced Ricci focusing,
- low shear in their interiors, and
- expansion-dominated dynamics.

Let a photon trajectory of total affine length D intersect a sequence of such basin regions. The total redshift may be written schematically as

$$\ln(1+z) = \frac{1}{3} \int \Theta_{\text{eff}}(\lambda) d\lambda,$$

which we decompose into contributions from basin and non-basin segments:

$$\ln(1+z) = \frac{1}{3} \left[\int_{\text{basin}} \Theta_{\text{void}} d\lambda + \int_{\text{non-basin}} \Theta_{\text{env}} d\lambda \right].$$

The additional contribution associated with MDN-organized regions may be expressed as

$$\Delta z_{\text{MDN}} \sim \frac{1}{3} \int_{\text{basin}} \delta\Theta(\lambda) d\lambda,$$

where $\delta\Theta \equiv \Theta_{\text{void}} - \Theta_{\text{env}}$.

8.3 Leading-Order Scaling Law

Within well-developed basin interiors, the Raychaudhuri equation implies that

- shear is suppressed,
- Ricci focusing is reduced, and
- null expansion remains positive over extended affine intervals.

Under these conditions, Θ_{void} varies slowly along the path and may be approximated as an effective constant over a basin segment of affine length L_i . To leading order,

$$\Delta z_{\text{MDN}} \sim \frac{1}{3} \sum_i \Theta_{\text{void},i} L_i.$$

Writing $\Theta_{\text{void}} \sim 3H_{\text{void}}$, this becomes

$$\Delta z_{\text{MDN}} \sim \frac{1}{3} \sum_i H_{\text{void},i} L_i.$$

Introducing the total path length D , we obtain the dimensionless scaling form

$$\Delta z_{\text{MDN}} \sim \frac{1}{3} H D \sum_i \left(\frac{L_i}{D} \right) (1 + \delta_i),$$

where δ_i encodes the relative enhancement of expansion within basin i compared to the background.

This expression shows that the contribution is governed not by local extrema, but by the cumulative affine path length spent within low-focusing basin regions.

8.4 Dimensionless Observable Parameter

To isolate the MDN contribution from the baseline cosmological redshift, we define a dimensionless observable

$$\Xi_{\text{MDN}} \equiv \frac{\Delta z_{\text{obs}}}{z_{\text{FLRW}}},$$

where z_{FLRW} is the redshift inferred under a homogeneous model.

Using the scaling relation above, this yields

$$\Xi_{\text{MDN}} = \frac{1}{3} \sum_i \left(\frac{L_i}{D} \right) \delta_i.$$

This parameter has a simple interpretation:

Ξ_{MDN} measures the weighted exposure of a photon trajectory to void-node-organized basin structure.

Large values correspond to:

- longer traversal through basin interiors,
- deeper or more strongly evacuated regions, or
- a greater cumulative fraction of the path in void-dominated geometry.

8.5 Behavior at Large Distances

At sufficiently large distances, a line of sight samples many such basin structures. In this limit, statistical averaging leads to

$$\langle \Xi_{\text{MDN}} \rangle \rightarrow \text{constant},$$

determined by the typical void fraction and basin depth. This mean contribution is observationally degenerate with homogeneous cosmological parameters and is expected to be absorbed into standard fits.

However, the effect does not vanish. The observable signature lies in the *fluctuations*

$$\delta \Xi_{\text{MDN}} = \Xi_{\text{MDN}} - \langle \Xi_{\text{MDN}} \rangle,$$

which remain sensitive to the specific line-of-sight structure. Different trajectories sample different sequences of basin regions, leading to measurable variance even when the mean is isotropic.

8.6 Observational Implication

The scaling relation implies a direct, testable statement:

At fixed distance, redshift residuals should correlate with the integrated void-node basin structure along the line of sight.

This correlation arises purely from geometric considerations and does not depend on a specific metric realization. The present work does not attempt to evaluate the magnitude of this effect in data, but identifies the relevant observable quantity for future investigation.

8.7 Status of the Result

The expressions derived in this section are leading-order scaling relations, not precision predictions. They establish that:

- the MDN contribution is path-extensive,
- it scales with basin exposure rather than local extrema, and
- it admits a natural dimensionless observable.

A quantitative evaluation requires explicit spacetime models and numerical ray tracing, which are beyond the scope of the present paper.

9. Assumptions and Scope

This analysis makes several controlled assumptions:

- photons are treated as test particles,
- shear is retained formally but treated as sub-leading,
- no explicit spacetime metric is constructed.

The key assumption that the effective expansion sampled by the null path is operationally meaningful when anchored to endpoint observables is stated explicitly. It motivates metric-level development, not modification of Einstein's equations.

10. Conclusion

Starting from the exact general-relativistic redshift formula, we have shown that when null and timelike congruences probe systematically different expansion histories, as occurs in spacetime organized around curvature minima, a non-canceling, path-integrated contribution to redshift arises naturally within classical general relativity.

In addition to establishing the existence of a path-integrated redshift contribution consistent with relativistic optics, the present analysis identifies its leading-order scaling behavior and a corresponding dimensionless observable quantity. The contribution scales with the cumulative affine path length through expansion-dominated, low-focusing basin regions organized by curvature minima, rather than being tied to localized extremal points. This distinction is essential: the effect is fundamentally path-extensive and therefore depends on the integrated geometric composition of the line of sight.

At sufficiently large distances, where many such basins are traversed, the mean contribution approaches statistical isotropy and becomes observationally degenerate with homogeneous cosmological parameters. However, the effect does not vanish. Fluctuations around this mean persist and encode the specific structure encountered along each null trajectory. Consequently, the primary observational signature is not a universal shift, but a structured variance: at fixed distance, redshift residuals are expected to correlate with the integrated void-node basin geometry along the line of sight. This identifies a concrete pathway for empirical investigation while remaining fully consistent with the geometric framework developed here.

The redshift effect presented is geometric rather than potential-based, path-extensive rather than boundary-localized, and best understood as an optical consequence of Curvature Extremum Duality. Whether this contribution plays a measurable cosmological role is an empirical question. The present work establishes only that the mechanism is consistent with relativistic optics and follows directly from congruence geometry.

Appendix A

Derivation of the Path-Integrated Redshift Ansatz from Congruence Geometry

A.1 Purpose and Status of This Appendix

This appendix provides a general-relativistic derivation of the phenomenological redshift accumulation ansatz employed in the main text. Its purpose is to demonstrate explicitly how a non-canceling, path-integrated contribution to cosmological redshift arises within classical general relativity when null and timelike congruences probe systematically different expansion histories.

The geometric setting for this analysis is established in Paper 1, where Mass Deficit Nodes (MDNs) as local minima of spacetime curvature organizing large cosmic voids are identified as geometric structures complementary to mass-concentration nodes (e.g., black holes) under Curvature Extremum Duality. In spacetime organized around curvature minima (“void nodes”), null geodesics traverse extended regions of expansion-dominated geometry that timelike worldlines do not access.

The derivation presented here:

- introduces no modification of Einstein’s equations,
- relies only on exact properties of null and timelike congruences,
- and makes all assumptions explicit.

No specific spacetime metric is constructed. The analysis is carried out entirely at the level of congruence geometry, which is sufficient to establish the existence, sign, and path-extensive nature of the effect. Metric-level realizations are deferred to subsequent work.

A.2 Exact Redshift Formula in General Relativity

The frequency ω of a photon measured by an observer with four-velocity u^a is given exactly by

$$\omega = -u^a k_a,$$

where k^a is the tangent vector to the photon’s null geodesic.

For a photon exchanged between an emitter and an observer, the observed redshift is therefore

$$1 + z = \frac{(k^a u_a)_{\text{em}}}{(k^a u_a)_{\text{obs}}}.$$

Taking the logarithmic derivative of $1 + z$ along the null geodesic parameterized by affine parameter λ yields

$$\frac{d}{d\lambda} \ln(1 + z) = -\frac{1}{k^a u_a} \frac{d}{d\lambda} (k^b u_b).$$

Using the geodesic equation $k^b \nabla_b k^a = 0$, this becomes the exact identity

$$\frac{d}{d\lambda} \ln(1 + z) = -\frac{1}{k^a u_a} k^b \nabla_b u_a.$$

No approximation has been made. Redshift in general relativity is governed by the interaction between the null geodesic and the observer congruence.

A.3 Decomposition of the Observer Congruence

The covariant derivative of the observer four-velocity admits the standard decomposition

$$\nabla_a u_b = \frac{1}{3} \Theta h_{ab} + \sigma_{ab} + \omega_{ab} - a_b u_a,$$

where:

- $\Theta = \nabla_a u^a$ is the volumetric expansion,
- σ_{ab} is the shear tensor,
- ω_{ab} is the vorticity tensor,
- $a^a = u^b \nabla_b u^a$ is the four-acceleration,
- $h_{ab} = g_{ab} + u_a u_b$ projects orthogonal to u^a .

Contracting this expression with the null tangent k^a and substituting into the redshift rate gives

$$\frac{d}{d\lambda} \ln(1 + z) = \frac{1}{3} \Theta_{\text{eff}}(\lambda) + \frac{k^a k^b \sigma_{ab}}{k^c u_c} + (\text{vorticity and acceleration terms}).$$

For cosmological observer congruences that are hypersurface-orthogonal and comoving with matter, vorticity vanishes and acceleration terms are negligible at large scales. The dominant contribution to the redshift rate is therefore governed by the effective expansion sampled along the null path, with shear entering as a correction.

A.4 Homogeneous Limit and Its Breakdown

In exact FLRW spacetime, the expansion scalar satisfies $\Theta = 3H$ everywhere. Null and timelike congruences probe identical expansion histories, reducing the redshift rate identically to

$$\frac{d}{d\lambda} \ln(1+z) = H,$$

which reproduces the standard cosmological redshift.

This cancellation relies on a geometric coincidence: null and timelike congruences must sample the same expansion history. In inhomogeneous spacetime, particularly in regions organized around curvature extrema, this coincidence need not hold.

A.5 Null–Timelike Congruence Decoupling in Curvature-Minimum Geometry

In spacetime organized around curvature minima (Mass Deficit Nodes / void nodes), photons propagate through extended regions of suppressed Ricci focusing in which null congruences exhibit persistent divergence. Timelike worldlines, by contrast, are dynamically confined to filamentary and wall structures associated with curvature maxima.

As a result, null and timelike congruences sample materially distinct expansion histories. This null–timelike congruence decoupling is the geometric origin of the effect analyzed here and arises naturally from Curvature Extremum Duality.

The relevant quantity governing redshift accumulation is therefore

$$\Theta_{\text{eff}}(\lambda) \equiv \Theta \text{ evaluated along the null geodesic.}$$

Substituting this into the redshift rate yields

$$\frac{d}{d\lambda} \ln(1+z) = \frac{1}{3} \Theta_{\text{eff}}(\lambda) + (\text{shear corrections}).$$

A.6 Path-Integrated Redshift Contribution

Integrating along the null path between emission and observation gives

$$\ln(1+z) = \frac{1}{3} \int_{\lambda_{\text{em}}}^{\lambda_{\text{obs}}} \Theta_{\text{eff}}(\lambda) d\lambda + (\text{sub-leading terms}).$$

Because $\Theta_{\text{eff}}(\lambda)$ remains elevated across extended void-node interiors, the contribution accumulates continuously with affine path length. There is no boundary-localized exit term that reverses this accumulation.

The contribution is therefore path-extensive, in contrast to potential-based effects.

A.7 Relation to ISW and Rees–Sciama Effects

Integrated Sachs–Wolfe and Rees–Sciama effects arise from explicit time derivatives of the gravitational potential along the photon path, schematically proportional to $\partial_t \Phi$. In static or slowly evolving curvature structure, these effects exhibit near-cancellation.

By contrast, the contribution derived here arises through the tensor $k^b \nabla_b u_a$ and depends on the geometric expansion of the congruence itself. It persists even for stationary curvature minima and does not rely on time-varying potentials.

The distinction is structural and tensorial, not interpretive.

A.8 Interpretation as Null-Averaged Optical Bias

The result may be interpreted as a null-averaged optical bias: null observables weight spacetime geometry by affine path length rather than spatial volume or mass distribution.

In strongly inhomogeneous spacetime, null averaging and spatial averaging are not equivalent. Because void-node-organized regions dominate affine path length, their geometric properties exert disproportionate influence on null observables even when global expansion is well approximated by FLRW.

A.9 Limitations and Assumptions

The derivation rests on the following controlled assumptions:

- photons are treated as test particles; backreaction diminishes as redshift accumulates,

- shear terms are retained but are sub-leading in large, approximately isotropic void interiors,
- the effective expansion sampled by the null path is operationally meaningful when anchored to endpoint observables.

These assumptions identify precisely where explicit metric-level analysis becomes necessary.

A.10 Summary

Starting from the exact general-relativistic redshift formula, we have shown that:

- redshift accumulation depends on the relative evolution of null and timelike congruences,
- when these congruences probe different expansion histories, as occurs in curvature-minimum (MDN / void-node) geometry, a path-integrated contribution arises naturally,
- the effect is geometric rather than potential-based and does not rely on evolving curvature,
- and it may be interpreted as a null-averaged optical consequence of Curvature Extremum Duality.

This appendix establishes the formal relativistic-optical foundation for the phenomenological accumulation law used in the main text and motivates subsequent metric-level and observational work.